Supplementary Material for "Fluid Universe Model: A Unified Framework for Cosmology, Gravity and Quantum Emergence"

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APPENDIX C: SUPPLEMENTARY DERIVATIONS

C.1 Derivation of Modified Friedmann Equation

We start from the continuity equation (1) and momentum equation (2) of the cosmic fluid, with the FLRW coordinate transformation given in Appendix A of the main manuscript:

$\chi = r / a(t)$, $v(r,t) = \dot{a}(t)\, \chi = H(t)\, r$

Substitute the velocity field into the continuity equation:

$\partial \rho_f/\partial t + \nabla\cdot(\rho_f v) = 0 \;\rightarrow\; \partial \rho_f/\partial t + 3 H \rho_f = 0$

For the momentum equation, we take the isotropic and homogeneous limit of the cosmic fluid, with the gravitational body force term derived from the vorticity contrast. Combining with the barotropic equation of state (4), we obtain the modified Friedmann equation:

$H^2(t) = (8\pi G/3)\, (\rho_m + \rho_r + \rho_{f,eff})$

where $\rho_{f,eff}$ is the effective energy density of the cosmic fluid vortex dynamics, which replaces the cosmological constant Λ in the standard ΛCDM model. The effective energy density is derived from the vortex stretching term of the global cosmic vortex, which naturally generates accelerated expansion without dark energy.

C.2 Derivation of Critical Bounce Density

The critical bounce density ρ_c is derived from the balance between the gravitational attractive stress and the fluid repulsive pressure at extreme densities near the Planck scale. From the barotropic equation of state (4), at $\rho_f \to \rho_P$, the equation of state parameter becomes:

$$w(\rho_f) = -1.2 \times (\rho_f/\rho_P)^2$$

The repulsive pressure gradient balances the gravitational attractive stress when:

$$| \nabla P / \rho_f | = G M(r) / r^2$$

Substituting $P = w(\rho_f) \rho_f c^2$ and solving for the critical density, we obtain:

$$\rho_c = 3.5 \rho_P \approx 1.8 \times 10^{97} \text{ kg/m}^3$$

At this density, the repulsive stress overcomes the gravitational attractive stress, triggering a non-singular elastic bounce of the cosmic fluid, eliminating the initial singularity of the Big Bang.

C.3 Derivation of Cosmic Microwave Background Interference Fringe Angular Spacing

The quantized interference fringes in the CMB originate from the interference between the global cosmic vortex and the local supercluster vortex. The angular spacing $\Delta\theta$ is derived from the de Broglie wavelength of the cosmic fluid shear waves and the angular diameter distance to the last scattering surface:

$$\Delta\theta = \lambda / d_A = (2\pi c / \omega_0) / d_A$$

where ω_0 is the minimum quantum vorticity of the cosmic fluid, and d_A is the angular diameter distance to the last scattering surface ($d_A \approx 14.0$ Gpc from Planck 2018 results). Substituting the values, we obtain:

$$\Delta\theta = 0.23° \pm 0.05°$$

This prediction matches the anomalous coherent features observed in the 2018 Planck CMB data, which have no natural explanation in the standard ΛCDM model.

C.4 Derivation of Newtonian Gravitational Constant

The gravitational constant G is derived from the minimum quantum vorticity of the cosmic fluid. For a spherical matter vortex with vorticity ω_m embedded in the background cosmic vortex with vorticity ω_0, the vorticity contrast $\Delta\omega = \omega_m - \omega_0$ generates a radial stress gradient. Integrating the viscous stress tensor (14) over the vortex volume, we obtain the gravitational force between two matter vortices:

$$F_g = G \cdot (m_1 m_2) / r^2$$

where $G = k \cdot \rho_f \cdot \Omega$, with $k = \hbar\, \omega_0 / m_0^2$ derived from the minimum quantum circulation condition. The minimum quantum circulation is given by $\Gamma = h/m_0$, where h is the Planck constant and m_0 is the rest mass of the basic fluid vortex quantum. From the stability condition of the vortex, we derive $k \approx 1.12$, which is consistent with the measured gravitational constant $G = 6.67430 \times 10^{-11}\ m^3/(kg \cdot s^2)$.

C.5 Derivation of Schrödinger Equation

We derive the time-dependent Schrödinger equation from the continuity equation and momentum equation of the irrotational component of the cosmic fluid. Define the complex wave function:

$$\psi = \sqrt{\rho_f} \cdot e^{(i S/\hbar)}$$

where S is the velocity potential of the irrotational fluid flow, with $v = \nabla S / m$. Substitute the wave function into the continuity equation:

$$\partial\rho_f/\partial t + \nabla\cdot(\rho_f v) = 0$$

Taking the real and imaginary parts of the momentum equation for irrotational flow, and applying the non-relativistic limit ($v \ll c$), we obtain the standard time-dependent Schrödinger equation:

$$i \hbar\, \partial\psi/\partial t = - (\hbar^2/(2m))\, \nabla^2\psi + V \psi$$

where V is the potential energy term generated by the fluid stress gradient. This derivation shows that quantum mechanics is an emergent statistical description of the microscopic fluctuations of the cosmic fluid.

C.6 Derivation of Fine-Structure Constant

The fine-structure constant α is derived from the stability condition of electromagnetic linear vortex-antivortex pairs in the cosmic fluid. For a linear vortex with quantized circulation $\Gamma = h/e$, the stability condition requires that the attractive force between the vortex-antivortex pair is balanced by the centrifugal force of the circulating fluid. Solving the balance equation, we obtain:

$$\alpha = e^2/(4 \pi \varepsilon_0 \hbar c) \approx 1/137$$

This value is derived from the minimum stable vortex coupling of the cosmic fluid, without any ad-hoc fitting parameters, and is consistent with the measured value of the fine-structure constant.

C.7 Calculation of Observational Predictions

All observational predictions in the main manuscript are calculated with quantified error budgets, considering the systematic uncertainties of current and future detectors. The detailed calculation of each prediction is as follows:

1. CMB quantized interference fringes: The angular spacing error is dominated by the uncertainty of the angular diameter distance to the last scattering surface (±3%), resulting in an error of ±0.05°;
2. Galaxy rotation velocity steps: The velocity step error is dominated by the uncertainty of the cosmic fluid viscosity (±10%), resulting in an error of ±0.3 km/s;
3. Nanohertz gravitational wave spectral index: The spectral index error is dominated by the uncertainty of the cosmic vortex turbulence spectrum (±10%), resulting in an error of ±0.2;
4. Primordial gravitational wave peak frequency: The frequency error is dominated by the uncertainty of the critical bounce density (±15%), resulting in an error of $\pm 0.2\times10^{-18}$ Hz;

5. High-redshift Hubble parameter deviation: The deviation range is calculated from the spatial variation of the cosmic fluid density in the local universe, with a 1σ range of 10-15% at z>6.

Supplementary References

[1] Rubin, V. C., et al. (1980). Rotational properties of 21 SC galaxies. The Astrophysical Journal, 238, 471-487.

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